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Research article

Numerical investigation of singularly perturbed time lag parabolic differential-difference equations

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ABSTRACT

This paper deals with the numerical investigation of a singularly perturbed parabolic differential-difference equation with a time lag. The proposed method comprises the θ method ($\theta = 1$) and the non-standard finite difference methods for temporal and spatial variable discretization, respectively. Besides, the Richardson extrapolation technique is employed to boost the accuracy and order of convergence of the scheme. The applicability of the proposed approach is confirmed through some numerical examples. Furthermore, we have improved the convergence from almost first to almost second order.

1. Introduction

The partial differential equations in which the highest-order derivative is multiplied by a small positive parameter and contains at least a delay parameter (s) are known as the singularly perturbed delay partial differential equations (SPDPDEs). Such types of equations arise in the mathematical modeling of population dynamics [25], immune response [1], variational problem in control theory [2,3], HIV infection [4], activation of neuronal variability [26], biological oscillators [7], mathematical ecology [6], physiological process [5], evolutionary biology [30], the neuronal variability [29,24] and more others. To investigate the future behavior of such types of problems, it is necessary to find their numerical solutions. However, it is very hard to solve SPDPDEs analytically due to the existence of a multiscale character in the solutions. So, it is necessary to construct ε-uniform convergent numerical approaches that solve SPDPDEs with time lag and mixed shifts in space successfully. In this work, we examine singularly perturbed parabolic differential equations (SPPDEs) with a time delay and mixed shifts in space variables:

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$$\begin{cases} \frac{\partial y(z,r)}{\partial r} - \varepsilon \frac{\partial^2 y(z,r)}{\partial z^2} + \imath(z)\frac{\partial y(z,r)}{\partial z} + \lambda(z)y(z-\delta,r) + \rho(z)y(z,r) + \sigma(z)y(z+\eta,r) \\ = -\nu(z,r)y(z,r-\tau) + s(z,r), (z,r) \in \Lambda, \\ y(z,r) = K_B(z,r), (z,r) \in \mu_B, \\ y(z,r) = K_L(z,r), \forall \in \mu_L = \left\{(z,r) : -\delta \le z \le 0, r \in \overline{P}_r\right\}, \\ y(z,r) = K_R(z,r), \forall \in \mu_R = \left\{(z,r) : 1 \le z \le 1+\eta, r \in \overline{P}_r\right\}, \end{cases} \tag{1.1}$$

where $\Lambda = P_z \times P_r = (0,1) \times (0,T], \mu = \mu_L \cup \mu_B \cup \mu_R$, μ_L and μ_R are the left and the right sides of the rectangular domain Λ corresponding to $z = 0$ and $z = 1$, respectively. $\mu_B = [0,1] \times [-\tau, 0]$. $0 < \varepsilon \ll 1$ is a singular perturbation parameter, η is advance parameter, δ and τ are delay parameters in space and time direction, respectively and $\delta, \eta, \tau = o(\varepsilon), \delta, \eta, \tau > 0$. The functions $\imath(z), \lambda(z), \rho(z), \sigma(z), \nu(z,r), s(z,r), K_B(z,r)$, $K_L(z,r)$ and $K_R(z,r)$ on Λ, are assumed to be sufficiently smooth, bounded and independent of ε.

If $\tau = 0$ Eq. (1.1) becomes to SPPPDEs with mixed shifts [24]:

$$\begin{cases} \frac{\partial y(z,r)}{\partial r} - \varepsilon \frac{\partial^2 y(z,r)}{\partial z^2} + \imath(z)\frac{\partial y(z,r)}{\partial z} + \lambda(z)y(z-\delta,r) + \rho(z)y(z,r) + \sigma(z)y(z+\eta,r) \\ = -\nu(z,r)y(z,r) + s(z,r), (z,r) \in \Lambda, \\ y(z,0) = K_0(z), z \in P_z, \\ y(z,r) = K_L(z,r), \forall \in \mu_L = \left\{(z,r) : -\delta \le z \le 0, r \in \overline{P}_r\right\}, \\ y(z,r) = K_R(z,r), \forall \in \mu_R = \left\{(z,r) : 1 \le z \le 1+\eta, r \in \overline{P}_r\right\}, \end{cases} \tag{1.2}$$

and used to describe the time evolution trajectories of the membrane potential. When $\delta = \eta = 0$ Eq. (1.1) becomes to SPPPDEs with time delay [22]:

$$\begin{cases} \frac{\partial y(z,r)}{\partial r} - \varepsilon \frac{\partial^2 y(z,r)}{\partial z^2} + \imath(z)\frac{\partial y(z,r)}{\partial z} + \lambda(z)y(z-\delta,r) + \rho(z)y(z,r) + \sigma(z)y(z+\eta,r) \\ = -\nu(z,r)y(z,r-\tau) + s(z,r), (z,r) \in \Lambda, \\ y(z,r) = K_B(z,r), (z,r) \in \mu_B, \\ y(0,r) = K_L(0,r), r \in \overline{P}_r, \\ y(1,r) = K_R(1,r), r \in \overline{P}_r, \end{cases} \tag{1.3}$$

and used to describe a furnace used to process metal sheets. The singular perturbation and retard parameters in Eq. (1.1) make it challenging to obtain oscillation-free solutions using standard numerical techniques. Non-standard, robust numerical approaches, such as fitted mesh and fitted operator methods, are commonly used. For the details of these methods, one can see [26–28].

Several scholars have extensively investigated SPPDDEs with either space or time delays. For instance, the authors in [8] have suggested an extended cubic B-spline collocation method for SPPDDE of Eq. (1.2). The collocation cubic B-spline method is presented for solving SPPDDE of Eq. (1.2) by [9]. Daba and Duressa [10] have developed a robust computational scheme for singularly perturbed differential equations with delay and advanced parameters. A novel numerical approach is constructed for SPPDDE of Eq. (1.2) by [16]. The authors in [13] and [15] have suggested finite difference methods on piecewise uniform Shishkin mesh for SPPDDE of Eq. (1.2). Daba and Duressa [33] have developed a parameter-uniform numerical method for SPPDDEs with discontinuous coefficients and large delay. Various numerical schemes for singularly perturbed differential-difference equations are given in [37]. A new stable finite difference method is presented for SPPDDEa with a time delay by [17]. Negero and Duressa [21] have developed an efficient numerical technique for SPPDDE with a time lag of Eq. (1.3). A robust numerical scheme for solving two parametric SPPDDEs involving time delay is constructed by [20]. Different numerical schemes have been constructed to treat pseudo-parabolic equation with a time delay in [34–36]. A parameter uniform numerical methods have been suggested for solving quasi-linear second-order delay differential equation by [38,39].

Recently, [18] and [19] have presented robust numerical schemes for Eq. (1.1). Gonfa and Daba [32] have developed robust numerical method for solving Eq. (1.1). Nevertheless, there are very few robust numerical schemes available for solving SPPDDE with a time lag. So, to minimize the observed limitation, in this work, an efficient numerical method is developed for Eq. (1.1).

Notations: Throughout the paper, $\|\cdot\|_\infty$ is defined by $\|F\|_\infty = \sup_{(z,r)} |F(z,r)|$ denotes the standard supremum norm for a function B defined on some domain μ. N and M denote the number of mesh points in space (z) and time (r) directions, respectively. C denotes a generic positive constant independent of the ε and the mesh sizes.

1.1. Properties of the solutions of continuous problem

Since $\delta, \eta, \tau < \varepsilon$, the use of Taylor's series expansion of the terms containing $y(z-\delta,r), y(z+\eta,r)$ and $y(z,r-\tau)$ is valid [23]. Thus, the use of Taylor's series expansion of these terms in Eq. (1.1) gives

$$\begin{cases} y(z-\delta,r) = y(z,r) - \delta y_z(z,r) + \frac{\delta^2}{2} y_{zz}(z,r) + O(\delta^3), \\ y(z+\eta,r) = y(z,r) + \eta y_z(z,r) + \frac{\eta^2}{2} y_{zz}(z,r) + O(\eta^3), \\ y(z,r-\tau) = y(z,r) - \tau y_r(z,r) + O(\tau^2). \end{cases} \tag{1.4}$$

The substitution of Eq. (1.4) into Eq. (1.1) gives

$$\begin{cases} \pounds_{c_\varepsilon} y(z,r) = \zeta(z,r)\frac{\partial y(z,r)}{\partial r} - c_\varepsilon \frac{\partial^2 y(z,r)}{\partial z^2} + \vartheta(z)\frac{\partial y(z,r)}{\partial z} + q(z)y(z,r) = s(z,r), (z,r) \in \Lambda, \\ y(z,0) = K_B(z,0), z \in \overline{P}_z, \\ y(0,r) = K_L(1,r), y(1,r) = K_R(1,r), r \in \overline{P}_r, \end{cases} \tag{1.5}$$

where $\zeta(z,r) = (1 - \nu(z,r)\tau)$, $c_\varepsilon(z) = \left(\varepsilon - \lambda(z)\frac{\delta^2}{2} - \sigma(z)\frac{\eta^2}{2}\right)$,
$\vartheta(z) = (\iota(z) - \lambda(z)\delta + \sigma(z)\eta)$, $q(z) = (\lambda(z) + \rho(z) + \sigma(z) + \nu(z,r))$. Since the difference between Eq. (1.1) and Eq. (1.5) is order of $O\left(\delta^3, \eta^3, \tau^2\right)$, Eq. (1.1) and Eq. (1.5) are asymptotically equivalent.

Lemma 1.1. *If $y \in C^{2,1}(\overline{\Lambda})$, $y\,\iota_{\partial\Lambda} \geq 0$ and $\pounds_\varepsilon y\,\iota_{\Lambda} \geq 0$, then $y\,\iota_{\overline{\Lambda}} \geq 0$.*

Proof. proof is given in [19]. □

Lemma 1.2. *Let $y(z,r)$ be the solution of the problem (1.5), then we have*

$$\|y\| \leq a^{-1}\|s\| + \max\left\{\left|K_B(z,0)\right|, \max\left\{\left|K_L(0,r)\right|, \left|K_R(1,r)\right|\right\}\right\},$$

where $\|.\|$ is the L_∞ norm given by $\|y\| = \max_{(z,r)\in\overline{\Lambda}}|y(z,r)|$ and $q((z) \geq a > 0$.

Proof. proof is given in [32]. □

Lemma 1.3. *The bound on the spatial derivative of the solution $y(z,r)$ of Eq. (1.5) and $\vartheta \geq \vartheta^* > 0$ is given by*

$$\left|\frac{\partial^k y(z,r)}{\partial z^k}\right| \leq C\left(1 + c_\varepsilon^{-k}\exp\left(\frac{-\vartheta^*(1-z)}{c_\varepsilon}\right)\right), (z,r) \in \overline{\Lambda}, k = 0(1)4.$$

Proof. proof is given in [9] □

2. Formulation of the numerical scheme

2.1. Semi-discretization in the temporal direction

The mesh for $P_r = [0,T]$ is given by:

$$P_r = \Lambda_{\Delta r}^M = \left\{r_j = j\Delta r, 0 \leq j \leq M, \Delta r = T/M\right\}.$$

Then, applying the $\theta-$ method ($\theta = 1$) on r of (1.5) yields

$$\begin{cases} \left(\zeta(z,r_{j+1}) + \Delta r \pounds_{c_\varepsilon}^M\right) Y(z,r_{j+1}) = -c_\varepsilon Y_{zz}(z,r_{j+1}) + \vartheta(z,r_{j+1})Y_z(z,r_{j+1}) + q(z,r_{j+1})Y(z,r_{j+1}) \\ -s(z,r_{j+1}) = \zeta(z,r_j)Y(z,r_j), \\ Y(z,0) = K_B(z,0), z \in \overline{P}_z, \\ Y(0,r_{j+1}) = K_L(0,r_{j+1}), Y(1,r_{j+1}) = K_R(1,r_{j+1}), 0 \leq j \leq M-1. \end{cases} \tag{2.1}$$

Lemma 2.1. *The local truncation error ($lte_{j+1} = y(x,t_{j+1}) - Y(x,t_{j+1})$) estimates with the semi-discretized of Eq. (2.1) is bounded by*

$$\left\|lte_{j+1}\right\|_\infty \leq C(\Delta r)^2.$$

Proof. Applying Taylor's series expansion to $y(z,r_{j+1})$ yields

$$y(z,r_{j+1}) = y(z,r_j) + \Delta r y_r(z,r_j) + O((\Delta r)^2).$$

This implies

$$\frac{y(z,r_{j+1}) - y(z,r_j)}{\Delta r} = y_r(z,r_j) + O(\Delta r). \tag{2.2}$$

The substitution of Eq. (1.5) into Eq. (2.2) yields

$$\frac{y(z,r_{j+1}) - y(z,r_j)}{\Delta r} = -\left(-c_\varepsilon(z)y_{zz}(z,r_j) + \vartheta(z,r_j)y_z(z,r_j) + q(z,r_j)y(z,r_j) - s(z,r_j)\right) + O(\Delta r). \tag{2.3}$$

This gives

$$\left(\zeta(z,r_{j+1}) + \Delta r \pounds_{c_\varepsilon}^M\right) y(z,r_{j+1}) - \Delta r s(z,r_j) + y(z,r_j) = O((\Delta r)^2). \tag{2.4}$$

Subtracting Eq. (2.1) from Eq. (2.4), we have

$$\begin{aligned}&\left(\zeta(z,r_{j+1})+\Delta r\pounds_{c_\varepsilon}^{M}\right)lte_{j+1}=O((\Delta r)^2),\\&lte_{j+1}(0)=0=lte_{j+1}(1).\end{aligned}\tag{2.5}$$

Hence, applying the maximum principle on the operator gives

$$\left\|lte_{j+1}\right\|_\infty\le C\,(\Delta r)^2\,.\quad\square$$

Lemma 2.2. *The global error estimate (gee) in the temporal direction of Eq.* (2.1) *satisfies*

$$\left\|gee_j\right\|_\infty\le C(\Delta r),\ \forall\, j\le T/\Delta r.$$

Proof. The gee is given by

$$\left\|gee_j\right\|_\infty\ =\ \left\|\textstyle\sum_{i=1}^{j}lte_i\right\|_\infty\le\left\|lte_1\right\|_\infty+\left\|lte_2\right\|_\infty+\left\|lte_3\right\|_\infty+\cdots+\left\|lte_j\right\|_\infty,\quad j\le T/\Delta r,\tag{2.6}$$

Using Lemma 2.1, we have

$$\left\|gee_j\right\|_\infty\le C\,(\Delta r)\,.\quad\square$$

2.1.1. Spatial discretization

We discretize Eq. (2.1) via a non-standard finite difference scheme on a uniform mesh. We divide the spatial domain $[0,1]$ into N equal parts with $\Delta z=1/N$:

$$P_z=\left\{z_0=0,z_i=i\Delta z,i=1(1)N-1,z_N=1\right\}.$$

For sake of simplicity, denote $Y(z,r_{j+1})\equiv Y(z)$, and from Eq. (2.1), we have:

$$-c_\varepsilon Y_{zz}(z)+\vartheta(z)Y_z(z)+Q(z)Y(z)=\phi(z),\tag{2.7}$$

where $Y(z)=Y^{j+1}(z),Q(z)=Q^{j+1}(z)=q^{j+1}(z)+\frac{\zeta^{j+1}(z)}{\Delta r}$ and $\phi(z)=s^{j+1}(z)+\zeta^{j+1}(z)\frac{Y^j(z)}{\Delta r}$.
We consider a homogeneous form of the constant coefficient sub-equation from Eq. (2.7) according to the procedures in [11,12] as:

$$-c_\varepsilon\left(\frac{Y_{i-1}-2Y_i+Y_{i+1}}{\gamma_i^2}\right)+\varphi\left(\frac{Y_{i+1}-Y_i}{\Delta z}\right)+\chi Y_i=0,\tag{2.8}$$

where $\varphi=\vartheta_i$ and $\chi=Q_i$.
The analytical solutions of Eq. (2.8) are: $\exp(\lambda_1 z)$ and $\exp(\lambda_2 z)$, where

$$\lambda_{1,2}=\frac{\varphi\pm\sqrt{\varphi^2+4c_\varepsilon\chi}}{2c_\varepsilon}\tag{2.9}$$

Using the concepts in [11]), we get

$$\begin{vmatrix}Y_{i-1}&\exp(\lambda_1 z_{i-1})&\exp(\lambda_2 z_{i-1})\\Y_i&\exp(\lambda_1 z_i)&\exp(\lambda_2 z_i)\\Y_{i+1}&\exp(\lambda_1 z_{i+1})&\exp(\lambda_2 z_{i+1})\end{vmatrix}=0.$$

After simplification, we have

$$-\exp(\frac{\varphi\Delta x}{2c_\varepsilon})Y_{i+1}+2\cosh\left(\frac{\Delta z\sqrt{\varphi^2+4c_\varepsilon\chi}}{2c_\varepsilon}\right)Y_i-\exp(\frac{-\varphi\Delta z}{2c_\varepsilon})Y_{i-1}=0,\tag{2.10}$$

which is the exact difference scheme for (2.8). After some manipulation of Eq. (2.10), we obtain the consistent scheme:

$$Y_{i+1}-2Y_i+Y_{i-1}+\left(\exp(\frac{\varphi\Delta z}{c_\varepsilon})-1\right)(Y_{i+1}-Y_i)=0.\tag{2.11}$$

From (2.11), we have

$$\varepsilon\frac{Y_{i-1}-2Y_i+Y_{i+1}}{\gamma^2}+\varphi\frac{Y_{i+1}-Y_i}{\Delta z}=0,\tag{2.12}$$

where

$$\gamma^2\equiv\gamma^2(\Delta z,c_\varepsilon)=\frac{\Delta z c_\varepsilon}{\varphi}\left(\exp(\frac{\varphi\Delta z}{c_\varepsilon})-1\right).\tag{2.13}$$

Adopting γ_i^2 to variable coefficients, we have:

$$\gamma_i^2(\Delta z, c_\varepsilon) \equiv \gamma_i^2 = \frac{\Delta z c_\varepsilon}{\vartheta_i}\left(\exp(\frac{\vartheta_i \Delta z}{c_\varepsilon}) - 1\right). \tag{2.14}$$

Applying the non-standard finite difference method of [12] on (2.7), we obtain:

$$\mathcal{L}_{c_\varepsilon}^{N,M} Y_i \equiv -c_\varepsilon \frac{Y_{i-1} - 2Y_i + Y_{i+1}}{\gamma_i^2} + \vartheta_i\left(\frac{Y_i - Y_{i-1}}{\Delta z}\right) + Q_i Y_i = \phi_i, \tag{2.15}$$

The discrete scheme in (2.15) takes the form:

$$AY = H, \quad 1 \le i \le N-1, \quad 0 \le j \le M, \tag{2.16}$$

$$\begin{cases} A_{i,i-1} = -\frac{c_\varepsilon}{\gamma_i^2} - \frac{\vartheta_i}{\Delta z}, \\ A_{i,i} = (\frac{2c_\varepsilon}{\gamma_i^2} + \frac{\vartheta_i}{\Delta z} + Q_i), \\ A_{i,i+1} = -\frac{c_\varepsilon}{\gamma_i^2}, \\ H_i = \phi_i. \end{cases} \tag{2.17}$$

Equation (2.16) is a tridiagonal matrix that is diagonally dominant and invertible. Therefore, the matrix A can be easily solved by the Thomas algorithm.

3. Uniform convergence analysis

Lemma 3.1. *If for any mesh function Υ_i^N, $\Upsilon_i^N \mid_{\partial\Lambda} \ge 0$ and $L_{c_\varepsilon}^{N,M}\Upsilon_i^N \mid_{\partial\Lambda} \ge 0$, for $i = 1(1)N-1$, then $\Upsilon_i^N \ge 0$, $i = 0(1)N$.*

Proof. Assume there exist i^* be such that $\Upsilon(z_{i^*}) = \min_{0\le i\le N} \Upsilon(z_i) < 0$. From the hypothesis, we have $i^* \ne 0, i^* \ne N$. Thus, we have

$$\mathcal{L}_{c_\varepsilon}^{N,M}\Upsilon_{i^*}^N \equiv -c_\varepsilon\left(\frac{\Upsilon_{i^*-1}^N - 2\Upsilon_{i^*}^N + \Upsilon_{i^*+1}^N}{\gamma}^2_{i^*}\right) + \vartheta_i^*\left(\frac{\Upsilon_{i^*}^N - \Upsilon_{i^*-1}^N}{\Delta z}\right) + Q_i^*\Upsilon_{i^*}^N < 0,$$

which contradicts the supposition, and hence, $\Upsilon_i^N \ge 0$ for all $i = 0(1)N$. □

Lemma 3.2. *If Υ_i^N is any mesh function such that $\Upsilon_i^N = \Upsilon_N^N = 0$, then*

$$|\Upsilon_i^N| \le \beta^{-1} \max_{1\le i\le N-1} |\mathcal{L}_{c_\varepsilon}^{N,M}\Upsilon_i^N|, \quad i = 0(1)N.$$

Proof. Let's define the two barrier functions as

$$\mathfrak{T}_i^{\pm} = \beta^{-1} \max_{i=1(1)N-1} |\mathcal{L}_{c_\varepsilon}^{N,M}\Upsilon_i^N| \pm \Upsilon_i^N.$$

Then

$$\mathfrak{T}^{\pm}(0) = \beta^{-1} \max_{i=1(1)N-1} |\mathcal{L}_{c_\varepsilon}^{N,M}\Upsilon_i^N| \pm \Upsilon^N(0) \ge 0.$$

$$\mathfrak{T}^{\pm}(1) = \beta^{-1} \max_{1\le i\le N-1} |\mathcal{L}_{c_\varepsilon}^{N,M}\Upsilon_i^N| \pm \Upsilon^N(1) \ge 0.$$

For $i = 1(1)N-1$,

$$\begin{aligned} \mathcal{L}_{c_\varepsilon}^{N,M}\mathfrak{T}_i^{\pm} &= \mathcal{L}_{c_\varepsilon}^{N,M}\left(\beta^{-1} \max_{i=1(1)N-1} |\mathcal{L}_{c_\varepsilon}^{N,M}\Upsilon_i^N| \pm \Upsilon_i^N\right) \\ &= \frac{Q_i}{\beta} \max_{i=1(1)N-1} |\mathcal{L}_{c_\varepsilon}^{N,M}\Upsilon_i^N| \pm \mathcal{L}_{c_\varepsilon}^{N,M}\Upsilon_i^N. \\ &\ge \max_{i=1(1)N-1} |\mathcal{L}_{c_\varepsilon}^{N,M}\Upsilon_i^N| \pm \mathcal{L}_{c_\varepsilon}^{N,M}\Upsilon_i^N, \quad \text{since } Q_i \ge \beta, \\ &\ge 0. \end{aligned}$$

An application of Lemma 3.1 gives

$$|\Upsilon_i^N| \le \beta^{-1} \max_{i=1(1)N-1} |\mathcal{L}_{c_\varepsilon}^{N,M}\Upsilon_i^N|, \quad i = 0(1)N. \quad \square$$

Lemma 3.3. *Let $Y(z)$ be the solution of the Eq. (2.7) and Y_i be the solution of Eq. (2.15). Then, the error estimate is given by*

$$\left|Y(z_i)-Y_i\right| \le CN^{-1}.\ \ 0\le i\le N$$

Proof. The error estimate of the scheme (2.15) is:

$$\begin{aligned}\mathcal{L}_{c_\varepsilon}^{N,M}\left(Y(z_i)-Y_i\right) &= \left[-c_\varepsilon Y_{zz}(z_i)+\vartheta(z_i)Y_z(z_i)+Q(z_i)Y(z_i)\right]\\ &\quad -\left[\frac{-c_\varepsilon\Delta z^2}{\gamma^2}D^+D^-Y(z_i)+\vartheta(z_i)D^-Y(z_i)+Q(z_i)Y(z_i)\right],\\ &= c_\varepsilon\left(\frac{\Delta z^2}{\gamma^2}D^+D^- - \frac{d^2}{dz^2}\right)Y(z_i)+\vartheta(z_i)\left(\frac{d}{dz}-D^-\right)Y(z_i),\end{aligned}$$

where

$$D^+D^-Y(z_i)=\frac{Y(z_{i+1})-2Y(z_i)+Y(z_{i-1})}{(\Delta z)^2}.$$

This implies that

$$\begin{aligned}\left|\mathcal{L}_{c_\varepsilon}^{N,M}\left(Y(z_i)-Y_i\right)\right| &\le c_\varepsilon\left|\left(\frac{(\Delta z)^2}{\gamma^2}-1\right)D^+D^-Y(z_i)\right|+c_\varepsilon\left|\left(D^+D^- - \frac{d^2}{dz^2}\right)Y(z_i)\right|\\ &\quad +\left|\vartheta(z_i)\left(\frac{d}{dz}-D^-\right)Y(z_i)\right|\end{aligned} \tag{3.1}$$

Simplifying as

$$\begin{cases}\frac{(\Delta z)^2}{\gamma^2}-1=-\frac{\vartheta(\varsigma)\Delta z}{2c_\varepsilon+\vartheta(\varsigma)\Delta z},\\ \left(D^+D^- - \frac{d^2}{dz^2}\right)Y(z_i)=\frac{(\Delta z)^2}{16}\frac{d^4Y(\varsigma)}{dz^4},\\ \left(\frac{d}{dz}-D^-\right)Y(z_i)=\frac{\Delta z}{2}\frac{d^2Y(\varsigma)}{dz^2},\end{cases} \tag{3.2}$$

for some ς, $z_{i-1}\le\varsigma\le z_{i+1}$
From Eq. (3.2) and Eq. (3.1), we obtain

$$\left|\mathcal{L}_{c_\varepsilon}^{N,M}\left(Y(z_i)-Y_i\right)\right| \le c_\varepsilon\frac{\vartheta(\varsigma)\Delta z}{2c_\varepsilon+\vartheta(\varsigma)\Delta z}\left\|\frac{d^2Y(\varsigma)}{dz^2}\right\|+c_\varepsilon\frac{(\Delta z)^2}{16}\left\|\frac{d^2Y(\varsigma)}{dz^2}\right\|+\vartheta(z_i)\frac{\Delta z}{2}\left\|\frac{d^2Y(\varsigma)}{dz^2}\right\| \tag{3.3}$$

Using Lemma 3.2 and the boundedness of $\vartheta(x)$, we get

$$\begin{aligned}\left|\mathcal{L}_{c_\varepsilon}^{N,M}\left(Y(z_i)-Y_i\right)\right| &\le \frac{\Delta z}{2c_\varepsilon+\vartheta(\varsigma)\Delta z}Cc_\varepsilon\left(1+c_\varepsilon^{-2}\exp\left(\frac{-\theta(1-z)}{c_\varepsilon}\right)\right)\\ &\quad +C(\Delta z)^2c_\varepsilon\left(1+c_\varepsilon^{-4}\exp\left(\frac{-\theta(1-z)}{c_\varepsilon}\right)\right)+C\Delta z\left(1+c_\varepsilon^{-2}\exp\left(\frac{-\theta(1-z)}{c_\varepsilon}\right)\right),\\ &\le C\frac{\Delta z2c_\varepsilon}{2c_\varepsilon+\theta\Delta z}\left(1+c_\varepsilon^{-2}\exp\left(\frac{-\theta(1-z)}{c_\varepsilon}\right)\right)+C(\Delta z)^2\left(c_\varepsilon+c_\varepsilon^{-3}\exp\left(\frac{-\theta(1-z)}{c_\varepsilon}\right)\right)\\ &\quad +C\Delta z\left(1+c_\varepsilon^{-2}\exp\left(\frac{-\theta(1-z)}{c_\varepsilon}\right)\right)\\ &\le C\frac{\Delta zc_\varepsilon}{2c_\varepsilon+\theta\Delta z}\left(1+c_\varepsilon^{-3}\exp\left(\frac{-\theta(1-z)}{c_\varepsilon}\right)\right)+C(\Delta z)^2\left(c_\varepsilon+c_\varepsilon^{-3}\exp\left(\frac{-\theta(1-z)}{c_\varepsilon}\right)\right)\\ &\quad +C\Delta z\left(1+c_\varepsilon^{-3}\exp\left(\frac{-\theta(1-z)}{c_\varepsilon}\right)\right),\quad \text{since,}\quad c_{\varepsilon^{-2}}\le c_{\varepsilon^{-3}},\\ &\le C\left(\frac{\Delta zc_\varepsilon}{2c_\varepsilon+\theta\Delta z}+(\Delta z)^2+\Delta z\right)\left(1+c_\varepsilon^{-3}\exp\left(\frac{-\theta(1-z)}{c_\varepsilon}\right)\right),\\ &\le C\Delta z\left(1+c_\varepsilon^{-3}\exp\left(\frac{-\theta(1-z)}{c_\varepsilon}\right)\right),\quad \text{for } i=0,1,2,\cdots,N.\end{aligned}$$

Apply limit as c_ε approaches to zero, we have

$$\lim_{c_\varepsilon\to 0}\left|\mathcal{L}_{c_\varepsilon}^{N,M}\left(Y(z_i)-Y_i\right)\right|\le C\Delta z.\ \ 0\le i\le N$$

Lemma 3.2 leads to

$$\left|Y(z_i)-Y_i\right|\le C\Delta z.\ \ 0\le i\le N \quad \square$$

Theorem 3.4. *Let $y(z,r)$ and Y_i^{j+1} be the solution of the Eq. (1.1) and the approximation of the solution of Eq. (2.15). Then, the error estimate is given by:*

$$\left|y(z_i,r_{j+1})-Y_i^{j+1}\right|\le C\left(\Delta r+\Delta z\right).\ \ 0\le i=0(1)N,\ j=0(1)M-1.$$

Proof. Using Lemmas 2.2 and 3.3, we obtain the required estimate. $\square$

3.1. Richardson extrapolation

Assume $\Lambda^{N,M} \subset \Lambda^{2N,2M}$ where $\Lambda^{N,M}$ is the mesh obtained from the mesh intervals Δz and Δr, and $\Lambda^{2N,2M}$ is the mesh obtained by bisecting the mesh intervals by Δz and Δr. Let $\tilde{Y}_i$ be the numerical solution obtained with $\Lambda^{2N,2M}$. From Theorem 3.4, it is clear that for the mesh point $(z_i, r_{j+1}) \in \Lambda^{N,M}$

$$\left| y(z_i, r_{j+1}) - Y_i \right| \leq C(\Delta z + \Delta r) + R^{N,M}, \tag{3.4}$$

As $\tilde{z}_{i+1} - \tilde{z}_i = \tilde{\Delta} z = \frac{\Delta z}{2}$ for $\tilde{z}_i \in \Lambda^{2N}$ and $\tilde{r}_{j+1} - \tilde{r}_j = \tilde{\Delta} r = \frac{\Delta r}{2}$ for $\tilde{r}_{j+1} \in \Lambda^{2M}$. For $(\tilde{z}_i, \tilde{r}_{j+1}) \in \Lambda^{2N,2M}$, we have

$$\left| y(z_i, r_{j+1}) - \tilde{Y}_i \right| \leq C(\frac{\Delta z}{2} + \frac{\Delta r}{2}) + R^{2N,2M}, \tag{3.5}$$

where $R^{N,M}$ and $R^{2N,2M}$ are the remainder terms with the truncation error of $O((\Delta z)^2 + \Delta r^2)$. From the inequalities in (3.4) and (3.5), we have

$$\left| y(z_i, r_{j+1}) - (2\tilde{Y}_i - y(z_i, r_{j+1})) \right| \leq C((\Delta z)^2 + (\Delta r)^2),$$

which gives

$$(Y_i)_{extr} = 2\tilde{Y}_i - Y_i, \tag{3.6}$$

is also an extrapolated numerical solution.

Theorem 3.5. *Let $y(z_i, r_{j+1})$ and $(Y_i)_{extr}$ be the solution of Eq. (1.1) and the extrapolated solution, respectively. Then, the new error bound takes the form*

$$\left\| (Y_i)_{extr} - y(z_i, r_{j+1}) \right\| \leq C((\Delta r)^2 + (\Delta z)^2).$$

Proof. See [31]. □

4. Numerical examples, results and discussion

Some test examples have been considered to demonstrate the applicability of the proposed scheme. The exact solution for the considered example does not exist. Therefore, the maximum absolute errors are computed by the double mesh principle [14]:

$$E^{N,M}_{\varepsilon,\delta,\eta} = \max_{(z_i, r_{j+1}) \in \Lambda^{N,M}} \left| \left(Y^{N,M}(z_i, r_{j+1}) - Y^{2N,2M}(z_i, r_{j+1}) \right) \right|, \text{ (Before extrapolation (BE))},$$

$$(E^{N,M}_{\varepsilon,\delta\eta})extp = \max_{(z_i, r_{j+1}) \in \Lambda^{N,M}} \left| \left(Y^{N,M}_{extp}(z_i, r_{j+1}) - Y^{2N,2M}_{extp}(z_i, r_{j+1}) \right) \right|, \text{ (After extrapolation(AE))},$$

where $Y\left(z_i, r_{j+1}\right)$ and $Y_{extp}\left(z_i, r_{j+1}\right)$ denote the numerical solution obtained before and after extrapolation in $\Lambda^{N,M}$ with N and M mesh intervals in the spatial and the temporal directions, respectively. The corresponding rate of convergence is computed by

$$r^{N,M}_{\varepsilon,\delta,\eta} = \log_2 \left(E^{N,M}_{\varepsilon,\delta,\eta} / E^{2N,2M}_{\varepsilon,\delta,\eta} \right), \text{(BE)},$$
$$(r^{N,M}_{\varepsilon,\delta,\eta})extp = \log_2 \left((E^{N,M}_{\varepsilon,\delta,\eta})extp / (E^{2N,2M}_{\varepsilon,\delta,\eta})extp \right), \text{(AE)}.$$

The parameter-uniform maximum absolute error and uniform order of convergence are determined using

$$E^{N,M} = \max_{\varepsilon,\delta,\eta} E^{N,M}_{\varepsilon,\delta,\eta}, \text{(BE)},$$
$$E^{N,M}_{extp} = \max_{\varepsilon,\delta,\eta} E^{N,M}_{\varepsilon,\delta,\eta,extp}, \text{(AE)},$$
$$r^{N,M} = \log_2 \left(E^{N,M} / E^{2N,2M} \right), \text{(BE)},$$
$$r^{N,M}_{extp} = \log_2 \left(E^{N,M}_{extp} / E^{2N,2M}_{extp} \right), \text{(AE)}.$$

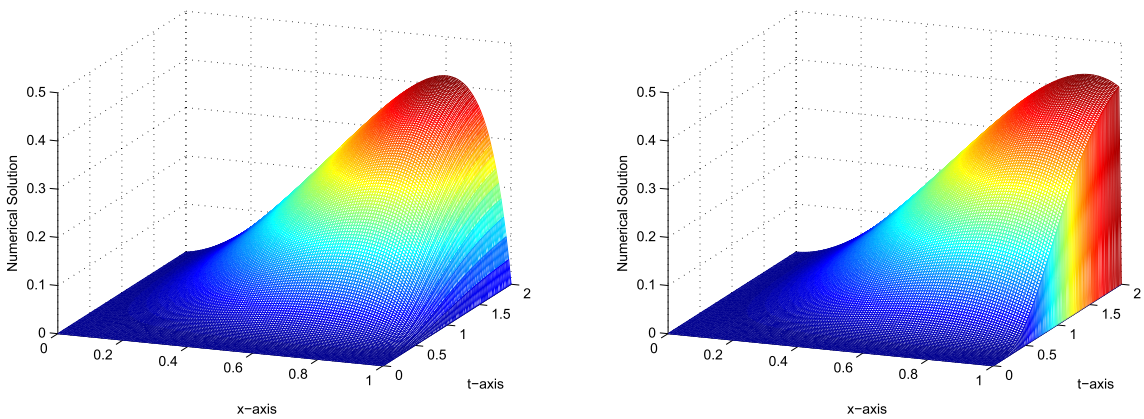
Example 4.1. Consider the following SPPDDE with time lag:

$$\begin{aligned} &\iota(z) = 2 - z^2, \lambda(z) = z - 3, s(z) = 1, \sigma(z) = 2, \upsilon(z, r) = 1 s(z, r) = 10 r^2 e^{(-r)} z(1 - z), \\ &(z, r) \in (0, 1) \times (0, 2], y(z, 0) = 0, z \in [0, 1], y(0, r) = 0 = y(1, r), r \in [0, 2]. \end{aligned}.$$

Tables 1 and 2 show that the proposed scheme has the first order of convergence before Richardson extrapolation and almost the second order of convergence after Richardson extrapolation, and the error has monotonically decreasing behavior with an increasing number of mesh intervals N and M, which supports the convergence of the presented method. A comparison of numerical results in Table 1 confirms that the proposed scheme provides more accurate results than the results in [18], [19], and [32]. Fig. 1 depicts

Table 1
Comparison of $E^{N,M}_{\varepsilon,\delta,\eta}$, $E^{N,M}$, $r^{N,M}$, $(E^{N,M}_{\varepsilon,\delta,\eta})_{extp}$, $E^{N,M}_{extp}$, and $r^{N,M}_{extp}$ for Example 4.1 using the presented scheme and results in [18,19,32] with $\delta = 0.1 \times \varepsilon, \eta = 0.1 \times \varepsilon, \tau = 0.5 \times \varepsilon$.

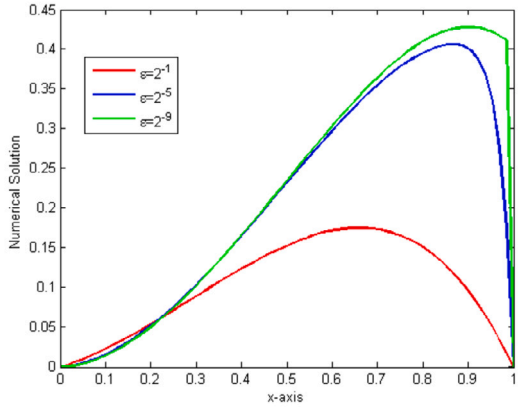
ε	N = 16	32	64	128	256	512
↓	M = 8	16	32	64	128	256
Before Richardson Extrapolation						
10^{-3}	1.2570e-02	7.6268e-03	4.1687e-03	2.1347e-03	9.3474e-04	4.4306e-04
10^{-5}	1.2577e-02	7.6305e-03	4.1716e-03	2.1753e-03	1.1100e-03	5.6058e-04
10^{-7}	1.2577e-02	7.6305e-03	4.1716e-03	2.1753e-03	1.1100e-03	5.6058e-04
10^{-9}	1.2577e-02	7.6305e-03	4.1716e-03	2.1753e-03	1.1100e-03	5.6058e-04
$E^{N,M}$	1.2577e-02	7.6305e-03	4.1716e-03	2.1753e-03	1.1100e-03	5.6058e-04
$r^{N,M}$	7.2094e-01	8.7118e-01	9.3939e-01	9.7065e-01	9.8557e-01	
After Richardson Extrapolation						
10^{-3}	1.8910e-03	8.4293e-04	3.6370e-04	1.3724e-04	4.677e-05	1.4901e-05
10^{-5}	1.8802e-03	8.4289e-04	3.6164e-04	1.3495e-04	4.6283e-05	1.4721e-05
10^{-7}	1.8802e-03	8.4289e-04	3.6164e-04	1.3495e-04	4.6283e-05	1.4721e-05
10^{-9}	1.8802e-03	8.4289e-04	3.6164e-04	1.3495e-04	4.6283e-05	1.4721e-05
$E^{N,M}_{extp}$	1.8910e-03	8.4293e-04	3.6370e-04	1.3724e-04	4.677e-05	1.4901e-05
$r^{N,M}_{extp}$	1.1657	1.2127	1.4060	1.5530	1.6502	
Results in [19]						
10^{-3}	2.9726e-2	1.4681e-2	5.0074e-3	1.6244e-3	5.1460e-4	1.5874e-4
10^{-5}	2.9660e-2	1.4702e-2	5.0373e-3	1.6758e-3	5.4486e-4	1.7267e-4
10^{-7}	2.9660e-2	1.4702e-2	5.0375e-3	1.6762e-3	5.4572e-4	1.7433e-4
10^{-9}	2.9660e-2	1.4702e-2	5.0375e-3	1.6762e-3	5.4573e-4	1.7434e-4
$E^{N,M}$	2.9726e-2	1.4681e-2	5.0375e-3	1.6762e-3	5.4573e-4	1.7434e-4
$r^{N,M}$	1.0178	1.5432	1.5875	1.6189	1.6463	
Results in [18]						
10^{-3}	4.1480e-2	2.8933e-2	2.0069e-2	1.3166e-2	8.1760e-3	4.8525e-4
10^{-5}	4.1687e-2	2.9042e-2	2.0150e-2	1.3216e-2	8.2042e-3	4.8694e-3
10^{-7}	4.1689e-2	2.9044e-2	2.0151e-2	1.3216e-2	8.2045e-3	4.8695e-3
10^{-9}	4.1689e-2	2.9044e-2	2.0151e-2	1.3216e-2	8.2045e-3	4.8695e-3
$E^{N,M}$	4.1689e-2	2.9044e-2	2.0151e-2	1.3216e-2	8.2045e-3	4.8695e-3
$r^{N,M}$	0.5214	0.5273	0.6085	0.6878	0.7526	
Results in [32]						
10^{-3}	8.6808e-3	5.5873e-3	3.1202e-3	1.6719e-3	8.3589e-4	4.5042e-4
10^{-5}	8.6852e-3	5.5901e-3	3.1225e-3	1.7041e-3	8.9350e-4	4.5725e-4
10^{-7}	8.6852e-3	5.5901e-3	3.1225e-3	1.7041e-3	8.9350e-4	4.5725e-4
10^{-9}	.6852e-3	5.5901e-3	3.1225e-3	1.7041e-3	8.9350e-4	4.5725e-4
$E^{N,M}$	8.6852e-3	5.5901e-3	3.1225e-3	1.7041e-3	8.9350e-4	4.5725e-4
$r^{N,M}$	6.3568e-1	8.4017e-1	8.7369e-1	9.3147e-1	9.6648e-1	



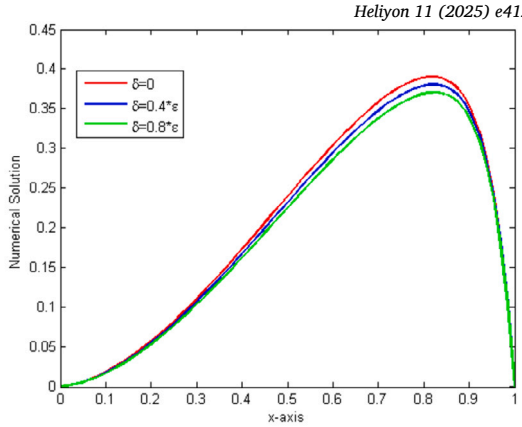
(a) $\varepsilon = 10^{-2}$ (b) $\varepsilon = 10^{-14}$

Fig. 1. Surface plot of numerical solutions with $N = M = 128, \delta = 0.5 \times \varepsilon, \eta = 0.6 \times \varepsilon, \tau = 0.5 \times \varepsilon$.

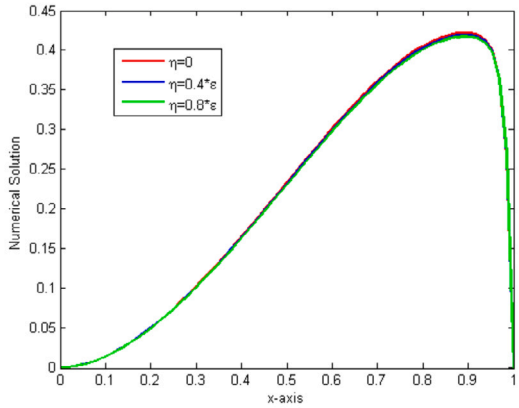
the interpretation of numerical data to support the theoretical findings about the right boundary layer. Fig. 2 (b) confirms that the width of the boundary layer increases as the values of δ increase, whereas the thickness of the layer decreases when the size of η increases (see Fig. 2 (c)). The proposed scheme's uniform convergence in a log-log scale plot, as depicted in Fig. 3, confirms the expected numerical order of convergence during rehearsal.



(a)When $\delta = 0.6 \times \varepsilon, \eta = 0.5 \times \varepsilon$ and different ε



(b) When $\varepsilon = 2^{-4}, \eta = 0.5 \times \varepsilon$ and different δ



(c) When $\varepsilon = 2^{-6}, \delta = 0.6 \times \varepsilon$ and different η

Fig. 2. Effect of the ε, δ and η on the solution profiles when $N = M = 64$.

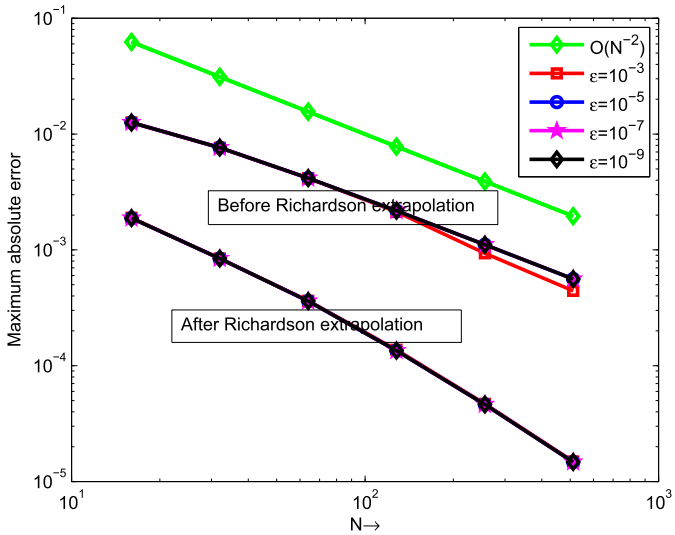


Fig. 3. Log-log plots for Example 4.1.

Table 2
$E^{N,M}_{\varepsilon,\delta,\eta}$, $E^{N,M}$, $r^{N,M}$, $(E^{N,M}_{\varepsilon,\delta,\eta})_{extp}$, $E^{N,M}_{extp}$, and $r^{N,M}_{extp}$ for Example 4.1 using the presented scheme with $\varepsilon = 10^{-2}$, $\tau = 0.5 \times \varepsilon$.

	N = 16	N = 32	64	128	256	512
↓	M = 8	16	32	64	128	256
Before Richardson Extrapolation						
$\delta \downarrow$	$\eta = 0.5 \times \varepsilon$					
$0.1 \times \varepsilon$	1.1888e-02	5.3545e-03	2.6023e-03	1.4981e-03	8.1029e-04	4.2334e-04
$0.2 \times \varepsilon$	1.1859e-02	5.3453e-03	2.5971e-03	1.4956e-03	8.0901e-04	4.2264e-04
$0.3 \times \varepsilon$	1.1830e-02	5.3353e-03	2.5920e-03	1.4930e-03	8.0770e-04	4.2193e-04
$0.4 \times \varepsilon$	1.1801e-02	5.3247e-03	2.5868e-03	1.4904e-03	8.0637e-04	4.2121e-04
$0.5 \times \varepsilon$	1.1771e-02	5.3133e-03	2.5817e-03	1.4878e-03	8.0503e-04	4.2048e-04
$E^{N,M}$	1.1888e-02	5.3545e-03	2.6023e-03	1.4981e-03	8.1029e-04	4.2334e-04
$r^{N,M}$	1.1507	1.0410	0.7967	0.8866	0.9366	
After Richardson Extrapolation						
$\delta \downarrow$	$\eta = 0.5 \times \varepsilon$					
$0.1 \times \varepsilon$	2.7048e-03	1.2631e-03	6.8235e-04	2.2100e-04	5.9300e-05	1.5105e-05
$0.2 \times \varepsilon$	2.6986e-03	1.2618e-03	6.8184e-04	2.2093e-04	5.9289e-05	1.5102e-05
$0.3 \times \varepsilon$	2.6928e-03	1.2602e-03	6.8125e-04	2.2081e-04	5.9261e-05	1.5095e-05
$0.4 \times \varepsilon$	2.6872e-03	1.2582e-03	6.8060e-04	2.2064e-04	5.9217e-05	1.5085e-05
$0.5 \times \varepsilon$	2.6820e-03	1.2560e-03	6.7987e-04	2.2041e-04	5.9158e-05	1.5069e-05
$E^{N,M}_{extp}$	2.7048e-03	1.2631e-03	6.8235e-04	2.2100e-04	5.9300e-05	1.5105e-05
$r^{N,M}_{extp}$	1.0986	0.8884	1.6265	1.8979	1.9730	
Before Richardson Extrapolation						
$\eta \downarrow$	$\delta = 0.5 \times \varepsilon$					
$0.1 \times \varepsilon$	1.1859e-02	5.3278e-03	2.5991e-03	1.4959e-03	8.0904e-04	4.2265e-04
$0.2 \times \varepsilon$	1.1836e-02	5.3231e-03	2.5948e-03	1.4939e-03	8.0803e-04	4.2209e-04
$0.3 \times \varepsilon$	1.1814e-02	5.3191e-03	2.5905e-03	1.4918e-03	8.0702e-04	4.2155e-04
$0.4 \times \varepsilon$	1.1792e-02	5.3159e-03	2.5861e-03	1.4898e-03	8.0602e-04	4.2101e-04
$0.5 \times \varepsilon$	1.1771e-02	5.3133e-03	2.5817e-03	1.4878e-03	8.0503e-04	4.2048e-04
$E^{N,M}$	1.1859e-02	5.3278e-03	2.5991e-03	1.4959e-03	8.0904e-04	4.2265e-04
$r^{N,M}$	1.1544	1.0355	0.7970	0.8867	0.9367	
After Richardson Extrapolation						
$\eta \downarrow$	$\delta = 0.5 \times \varepsilon$					
$0.1 \times \varepsilon$	2.7115e-03	1.2550e-03	6.8063e-04	2.2004e-04	5.9010e-05	1.5029e-05
$0.2 \times \varepsilon$	2.7046e-03	1.2548e-03	6.8036e-04	2.2008e-04	5.9030e-05	1.5034e-05
$0.3 \times \varepsilon$	2.6974e-03	1.2549e-03	6.8015e-04	2.2015e-04	5.9061e-05	1.5043e-05
$0.4 \times \varepsilon$	2.6899e-03	1.2553e-03	6.7998e-04	2.2026e-04	5.9104e-05	1.5055e-05
$0.5 \times \varepsilon$	2.6820e-03	1.2560e-03	6.7987e-04	2.2041e-04	5.9158e-05	1.5069e-05
$E^{N,M}_{extp}$	2.7115e-03	1.2560e-03	6.8063e-04	2.2041e-04	5.9158e-05	1.5069e-05
$r^{N,M}_{extp}$	1.1103	0.8839	1.6267	1.8975	1.9730	

5. Conclusion

An efficient uniform numerical scheme for singularly perturbed parabolic differential-difference with time delay is presented, using the θ-method in time and the non-standard method in space on uniform meshes. To enhance the convergence of the scheme, the Richardson extrapolation technique is used. The proposed scheme demonstrated better numerical results compared to other available numerical approaches in the literature. Moreover, we confirmed our theoretical error calculations through some numerical experiments, which depicted first-order convergence independent of the perturbation parameter (ε) before Richardson extrapolation and almost-second-order convergence independent of ε after Richardson extrapolation.

CRediT authorship contribution statement

Imiru Takele Daba: Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Wondwosen Gebeyaw Melesse:** Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Fasika Wondimu Gelu:** Writing – review & editing, Visualization, Validation, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Guta Demisu Kebede:** Writing – review & editing, Writing – original draft, Validation, Methodology, Formal analysis, Data curation, Conceptualization.

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Declaration of competing interest

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Data availability

Not applicable.

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Corrigendum

Corrigendum to "Numerical investigation of singularly perturbed time lag parabolic differential-difference equations" [Heliyon Volume 11, Issue 1, January 2025, Article e41215]

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In this article, references [24], [26] and [30] need to be corrected due to contextual inappropriateness:

[24] M. Musila, P. Lánsk'y, Generalized Stein's model for anatomically complex neurons, 1991.

[26] E.P. Doolan, J.J.H. Miller, H.A. Schilders W, Uniform Numerical Methods for Problems with Initial and Boundary Layers, Boole Press, 1980.

[30] M. Wazewska-Czyzewska, A. Lasota, Mathematical models of the red cell system, Mat. Stosow. 6 (1) (1976).

The authors confirm that the references were inadvertently introduced in the article and they should be considered as removed from the reference list and their intext citations ignored.

The ***authors*** apologize for the errors.

Declaration of competing interest

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